

Diagram 1: CFAR Sliding Window

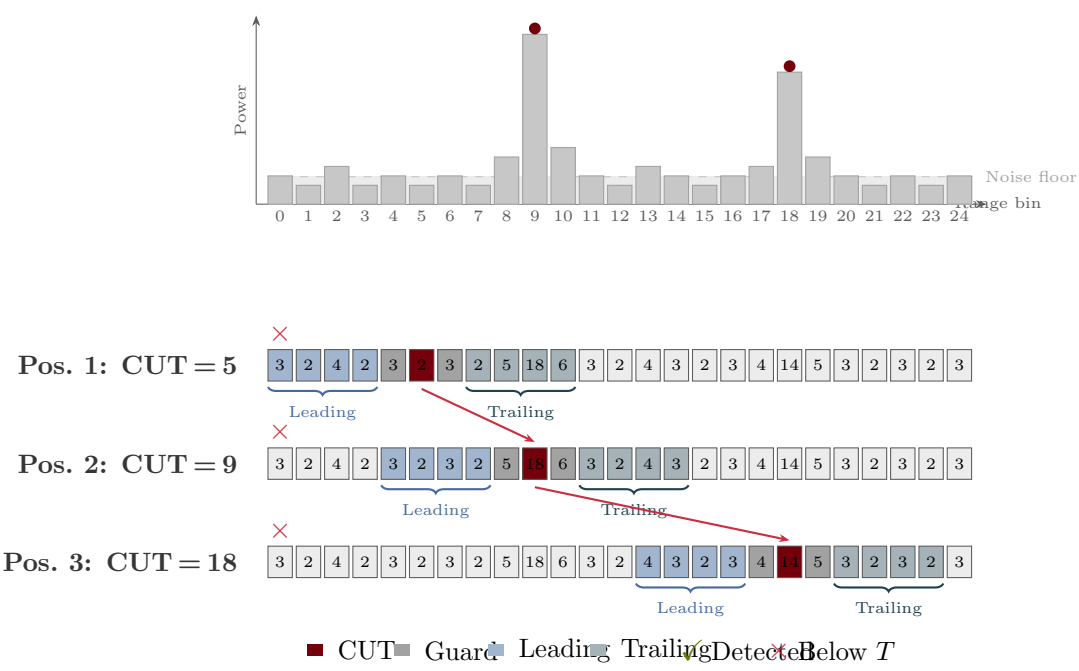
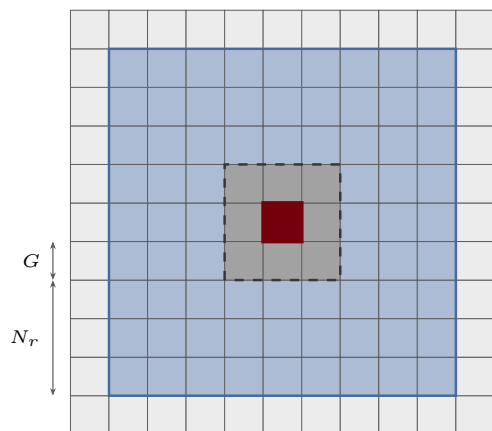


Diagram 1b: CFAR Variant Comparison

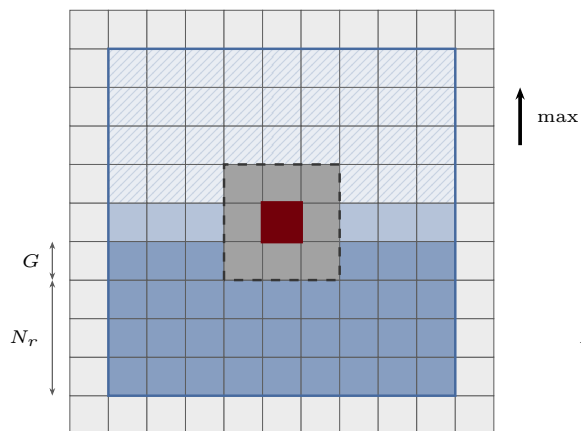
CA-CFAR



$$Z = \frac{1}{2N} \sum_{i=1}^{2N} x_i$$

Detect if $x_{\text{CUT}} > \alpha \cdot Z$

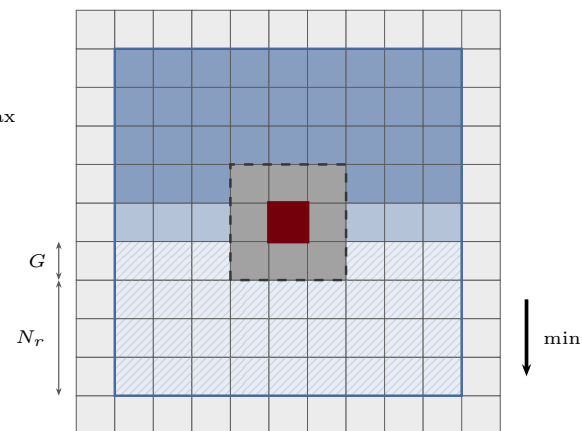
GO-CFAR



$$Z = \max(\bar{x}_L, \bar{x}_T)$$

Detect if $x_{\text{CUT}} > \alpha \cdot Z$

SO-CFAR



$$Z = \min(\bar{x}_L, \bar{x}_T)$$

Detect if $x_{\text{CUT}} > \alpha \cdot Z$

CUT
 Guard
 Selected Ref.
 Unused Ref.
 All Ref. (CA)

Diagram 2: Range-Azimuth to PPI Mapping

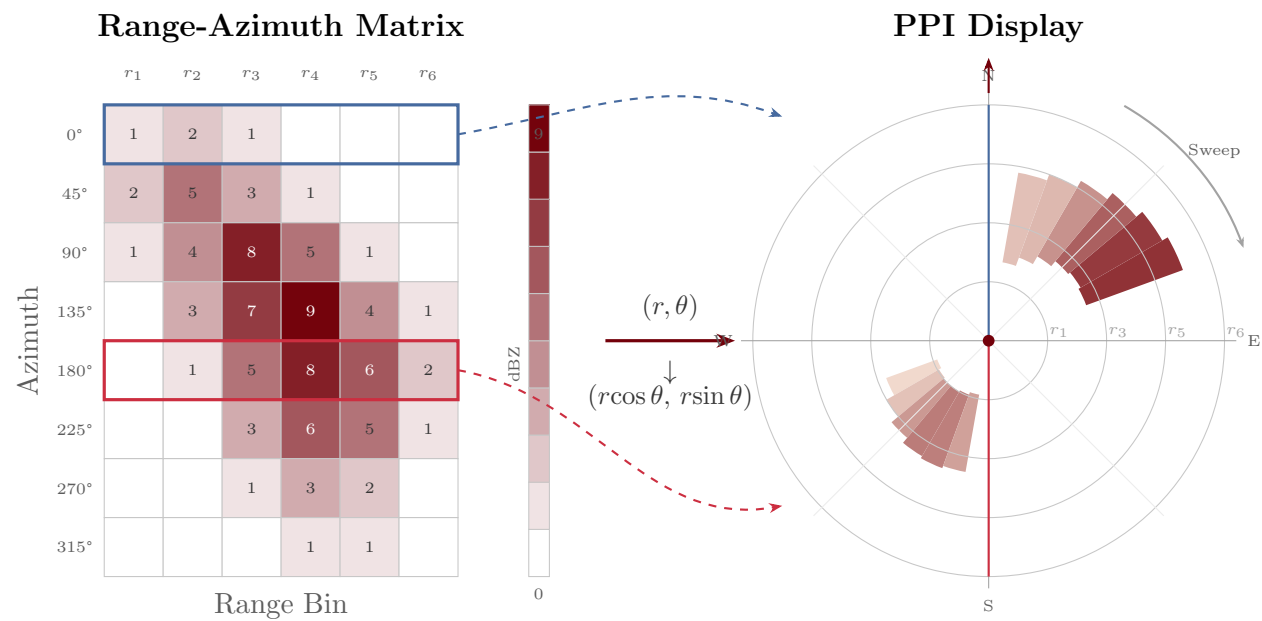


Diagram 3: Coverage Score Illustration

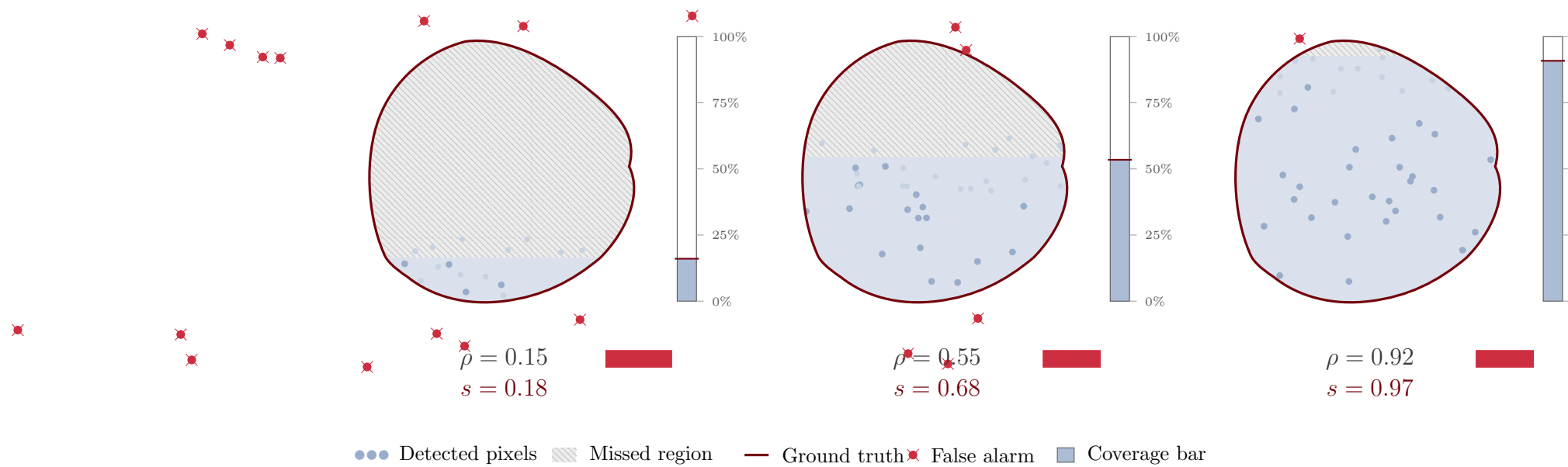


Diagram 4: Guard Cell Size Effect

(a) Guard Too Small

(b) Guard Appropriate

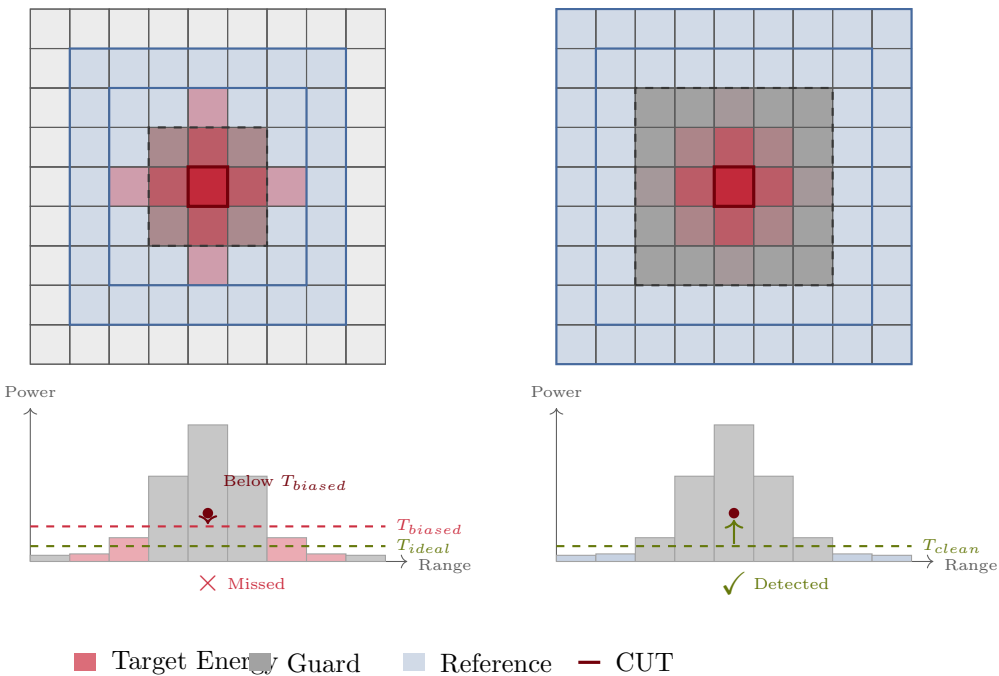


Diagram 5: Integral Image (Summed Area Table)

